

Attention-Augmented Retrieval over Long Documents with Compressed Section Memory Banks

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1 Introduction

Retrieval systems routinely discard document structure. We propose a hierarchical index whose sections carry their own vectors, and measure the effect of attention pooling on long-span representations across three corpora.

2 Related Work

Prior work on passage retrieval concentrates on fixed-size windows. Sentence-aware chunking preserves boundaries but ignores headings entirely.

3 Method

Our encoder reads each section independently, then pools window vectors by weighted mean. The pooling temperature is corpus-dependent and set by validation.

4 Results

The hierarchical index wins on section targets and ties on document targets, matching the intuition that structure helps precise queries most.